Sequential Four-Digit Door Lock with Attempt Lockout

NSCS204 Computer Organization and Architecture Lab · Department of Computer Science and Engineering, IIT (ISM) Dhanbad

Practical Digital Systems in Logisim, Project 2

Field	Entry
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Group	3
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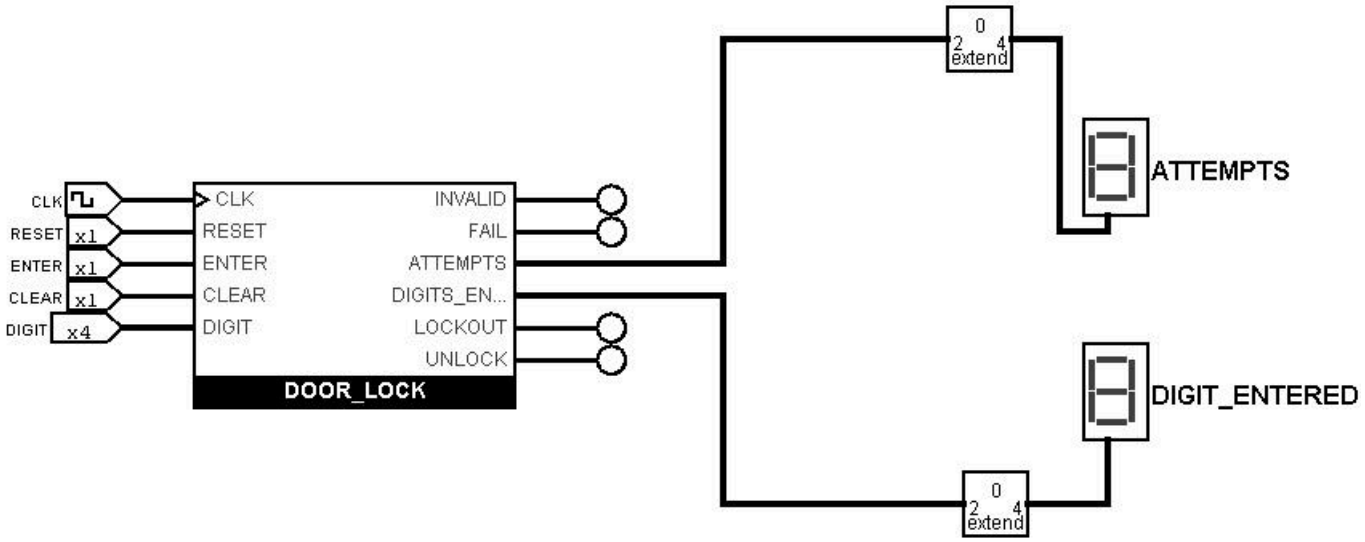
1. Problem Overview and Interfaces

The project is a door lock with the fixed PIN 2468. Decimal digits are entered one at a time. Four valid digits form one complete attempt. A correct attempt unlocks the door for exactly five seconds (320 master periods at $64\ \mathrm{Hz}$). Three consecutive wrong complete attempts cause a persistent lockout that only the administrator reset (RESET) can clear. Partial entries, invalid digits (10 to 15) and held buttons have defined behaviour, described in Section 3.

1.1 Interface

Signal	Direction	Bits	Meaning
CLK	input	1	Master clock, rising edge, $64\ \mathrm{Hz}$
RESET	input	1	Active HIGH, synchronous, highest priority
DIGIT	input	4	BCD digit, 0 to 9 valid, 10 to 15 invalid
ENTER	input	1	Press event: capture DIGIT
CLEAR	input	1	Press event: abandon the partial entry
UNLOCK	output	1	HIGH for exactly 320 periods after a correct PIN
LOCKOUT	output	1	HIGH permanently after the third wrong attempt
FAIL	output	1	Wrong-attempt indication
INVALID	output	1	One-period pulse for an invalid ENTER in ENTRY
ATTEMPTS	output	2	Wrong-attempt count 0 to 3, shown on a seven-segment display
DIGITS_ENTERED	output	2	Partial-entry length 0 to 3, shown on a seven-segment display

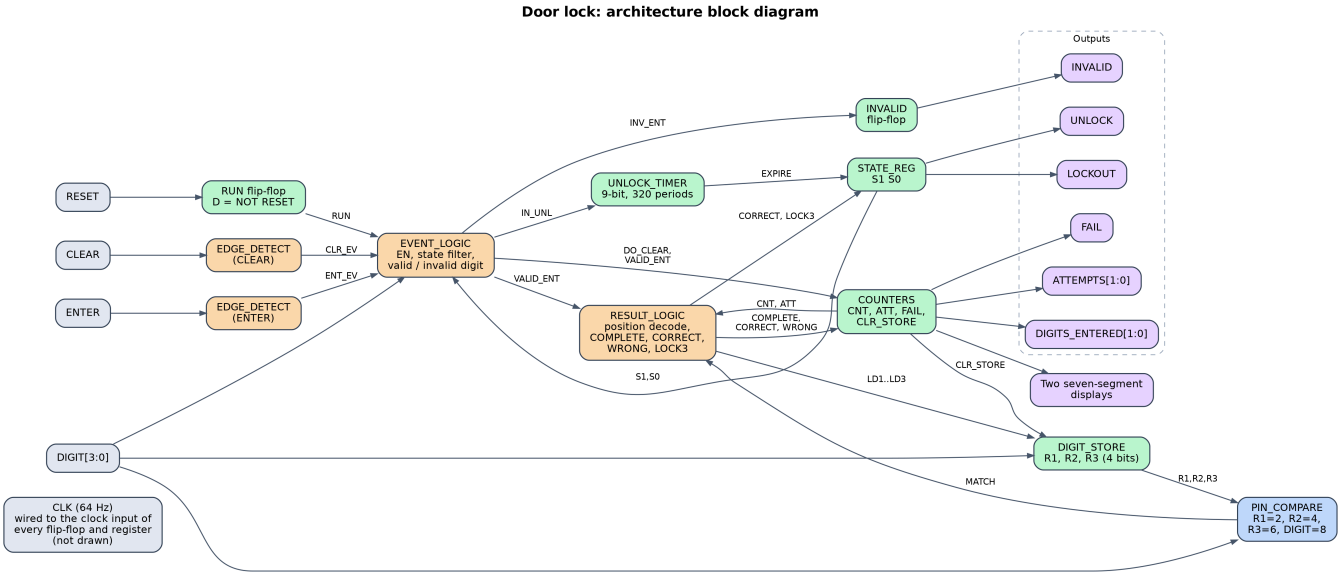
All button inputs are active HIGH. Reset state: `ENTRY`, empty digit storage, `ATTEMPTS=0`, `DIGITS_ENTERED=0`, all indicators LOW.



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2. Architecture and Component Choices

2.1 Block diagram



2.2 Subcircuits

Each block is built as a separate Logisim subcircuit and tested alone before assembly into `DOOR_LOCK` and the top-level `MAIN` wrapper.

Subcircuit	Type	Inputs	Outputs	Function
EDGE_DETECT	sequential	BTN, CLK	EV, PREV	One press event from a held button
EVENT_LOGIC	combinational	RUN, RESET, S1, S0, CLR_EV, ENT_EV, DIGIT[4]	EN, IN_ENTRY, IN_UNL, DO_CLEAR, VALID_ENT, INV_ENT	Event enabling, state filter, digit validity

Subcircuit	Type	Inputs	Outputs	Function
RESULT_LOGIC	combinational	CNT[2], VALID_ENT, MATCH, ATT[2]	C_IS0, C_IS3, LD1, LD2, LD3, COMPLETE, CORRECT, WRONG, LOCK3	Position decode and attempt result
PIN_COMPARE	combinational	R1, R2, R3, DIGIT (4 bits each)	MATCH	Equality with $2, 4, 6, 8$
DIGIT_STORE	sequential	CLR_STORE, DIGIT[4], LD1, LD2, CLK, LD3	R1, R2, R3 (4 bits each)	Captured first, second, and third digits
COUNTERS	sequential	RESET, CLK, VALID_ENT, C_IS3, CORRECT, WRONG, DO_CLEAR, COMPLETE, C_IS0	CNT[2], ATT[2], CLR_STORE, FAIL	Partial length, wrong attempts, FAIL
UNLOCK_TIMER	sequential	IN_UNL, RESET, CLK	TMR[9], EXPIRE	320-period timer
STATE_REG	sequential	CLK, RESET, CORRECT, LOCK3, EXPIRE, IN_UNL	S0, S1, UNLOCK, LOCKOUT	State register and decoded outputs
DOOR_LOCK	subcircuit	CLK, RESET, ENTER, CLEAR, DIGIT[4]	INVALID, FAIL, ATTEMPTS[2], DIGITS_ENTERED[2], LOCKOUT, UNLOCK	Core door lock assembly, RUN and INVALID flip-flops
MAIN	top level	CLK, RESET, ENTER, CLEAR, DIGIT[4]	LEDs (INVALID, FAIL, LOCKOUT, UNLOCK) and Hex Displays (ATTEMPTS, DIGIT_ENTERED)	Interactive top-level test harness with displays and indicators

2.3 Components and configuration

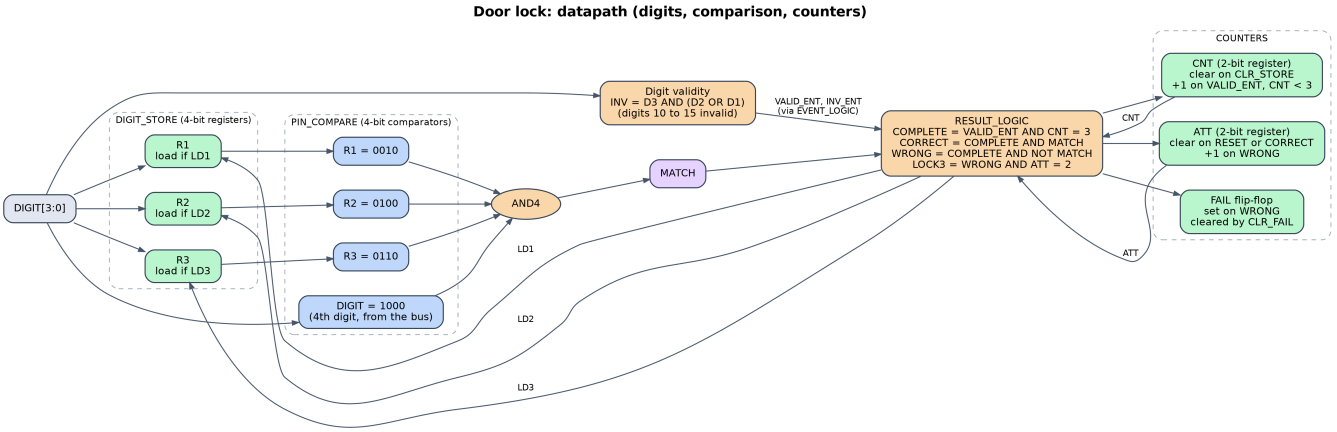
Component	Configuration	Where used
D Flip-Flop	Rising edge, asynchronous pins unused	PREV history, RUN, S1, S0, FAIL, INVALID
Register	Rising edge, enable explicit, clear pin tied to 0	R1 to R3 (4 bits), CNT and ATT (2 bits), TMR (9 bits)
Adder	2 bits and 9 bits	$\mathrm{CNT}+1$, $\mathrm{ATT}+1$, $\mathrm{TMR}+1$
Comparator	4 bits and 9 bits, equal output	PIN compare, $\mathrm{TMR}=319$
Multiplexer	$2:1$, 2 to 9 bits	Synchronous clearing
Decoder	2-to-4, no enable	Position decode of CNT
Bit Extender	2 bits to 4 bits, zero extension	Driving Hex Digit Displays in MAIN
Hex Digit Display	decimal point disabled (`false`)	ATTEMPTS and DIGIT_ENTERED

Component	Configuration	Where used
LED	Active HIGH (Yellow, Orange, Red, Green)	INVALID, FAIL, LOCKOUT, UNLOCK indicators in MAIN
Gates	AND, OR, NOR, NOT, with negate attributes	Control logic

All resets are synchronous and folded into the next-state logic. No asynchronous clear pin is used for `RESET`. No external program implements the controller.

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3. Controller and Datapath



3.1 Signal and storage widths

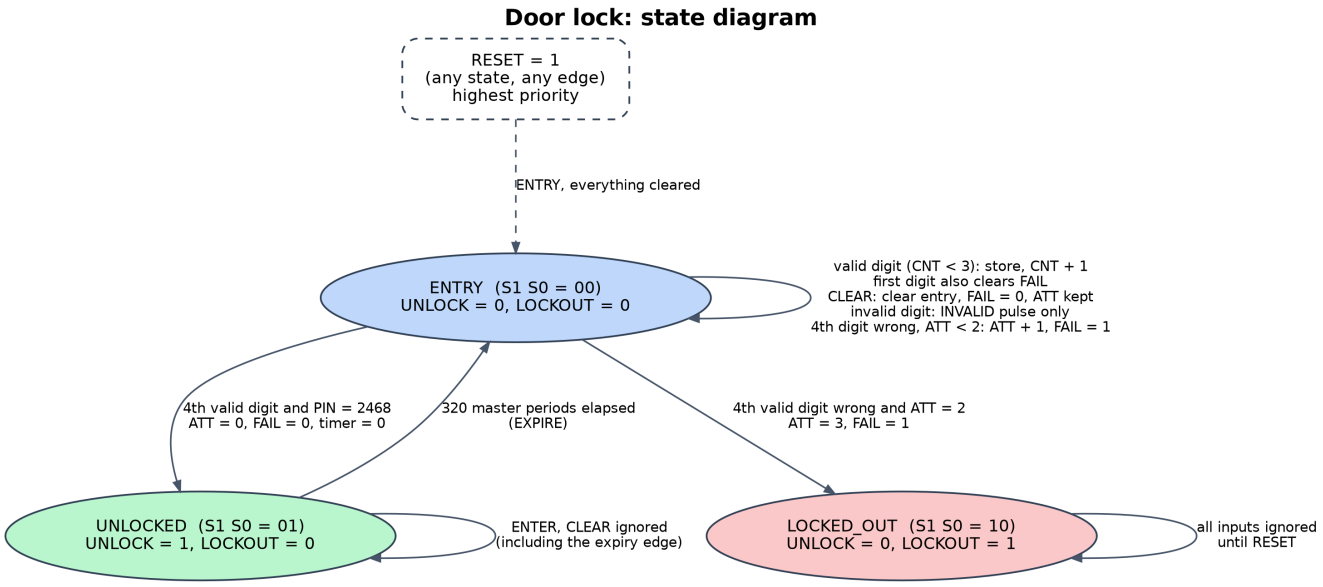
Storage	Width	Purpose
S (S1, S0)	2	State
RUN	1	Startup-boundary flag
ENTER_PREV, CLEAR_PREV	1 each	Button history (not reset)
R1, R2, R3	4 each	First, second and third captured digits
CNT (DIGITS_ENTERED)	2	Partial length 0 to 3
ATT (ATTEMPTS)	2	Wrong attempts 0 to 3
FAIL	1	Wrong-attempt flag
INVALID	1	Registered one-period pulse
TMR	9	Unlock timer, 0 to 319 ($2^9 = 512 > 319$)

The fourth digit is not stored. It is compared directly from the `DIGIT` bus on the edge on which it is accepted, which satisfies the requirement to compare including the digit just accepted.

3.2 State encoding and state diagram

State	S1 S0	UNLOCK	LOCKOUT
ENTRY	00	0	0

State	S1 S0	UNLOCK	LOCKOUT
UNLOCKED	01	1	0
LOCKED_OUT	10	0	1
unused	11	not reachable	



3.3 State transition table

Current	Condition	Next	Side effects
any	$\mathrm{RESET} = 1$	ENTRY	clear everything (highest priority)
ENTRY	CLEAR event	ENTRY	clear digits, $\mathrm{CNT} = 0$, $\mathrm{FAIL} = 0$, ATTEMPTS preserved
ENTRY	valid ENTER, $\mathrm{CNT} < 3$	ENTRY	store digit, $\mathrm{CNT} + 1$; the first digit also clears FAIL
ENTRY	invalid ENTER	ENTRY	INVALID pulse only
ENTRY	4th valid digit, PIN matches	UNLOCKED	$\mathrm{ATT} = 0$, $\mathrm{FAIL} = 0$, $\mathrm{CNT} = 0$, $\mathrm{timer} = 0$
ENTRY	4th valid digit, mismatch, $\mathrm{ATT} < 2$	ENTRY	$\mathrm{ATT} + 1$, $\mathrm{FAIL} = 1$, $\mathrm{CNT} = 0$
ENTRY	4th valid digit, mismatch, $\mathrm{ATT} = 2$	LOCKED_OUT	$\mathrm{ATT} = 3$, $\mathrm{FAIL} = 1$, $\mathrm{CNT} = 0$
UNLOCKED	timer reaches 320 periods	ENTRY	UNLOCK falls, entry empty
UNLOCKED	ENTER or CLEAR	UNLOCKED	ignored, including the expiry edge
LOCKED_OUT	anything except RESET	LOCKED_OUT	ignored

3.4 Input-event logic

Clock. One rising-edge master clock at $64\ \mathrm{Hz}$ (Section 4).

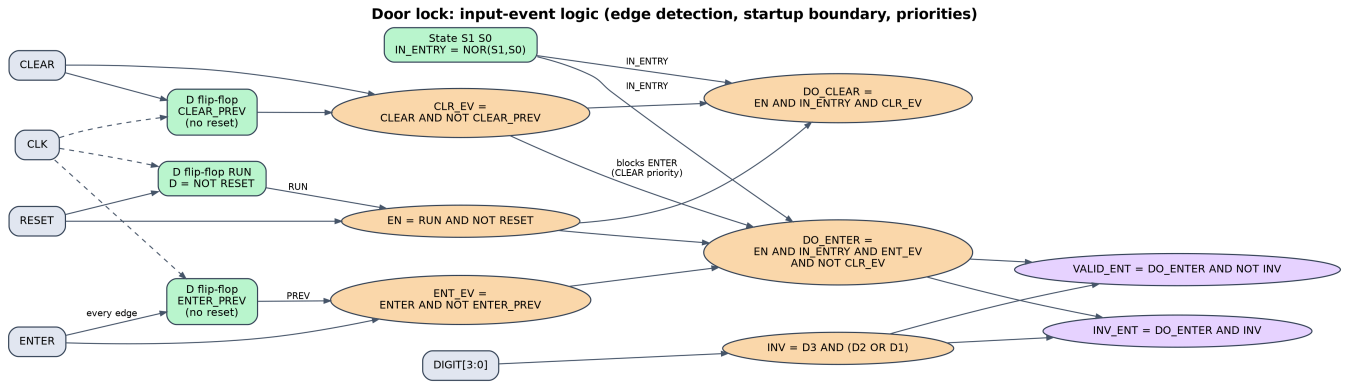
Button events. Each button has a history flip-flop PREV that samples the button on every rising edge. The event is BUTTON AND NOT PREV, evaluated from pre-edge values. The PREV flip-flops are not cleared by RESET and they keep updating while a command is ignored. Consequences: a held button gives one event only; a button held HIGH across reset must be released and pressed again; a button held through the unlock expiry cannot fire a deferred action.

Startup boundary. A flip-flop RUN has $D = \text{NOT RESET}$. While $\text{RESET} = 1$, $\text{RUN} = 0$. On the first edge after RESET falls, the pre-edge RUN is still 0, so no event is accepted, and RUN becomes 1. Events are accepted from the next edge.

Equations (all signals are pre-edge values):

```
EN        = RUN AND NOT RESET
ENT_EV    = ENTER AND NOT ENTER_PREV
CLR_EV    = CLEAR AND NOT CLEAR_PREV
IN_ENTRY  = NOT S1 AND NOT S0
IN_UNL    = NOT S1 AND S0
DO_CLEAR  = EN AND IN_ENTRY AND CLR_EV
DO_ENTER  = EN AND IN_ENTRY AND ENT_EV AND NOT CLR_EV
INV       = DIGIT3 AND (DIGIT2 OR DIGIT1)          ; digits 10 to 15
VALID_ENT = DO_ENTER AND NOT INV
INV_ENT   = DO_ENTER AND INV
LD1 = VALID_ENT AND (CNT == 0)    LD2 = VALID_ENT AND (CNT == 1)    LD3 = VALID_ENT AND
(CNT == 2)
COMPLETE  = VALID_ENT AND (CNT == 3)
MATCH     = (R1 == 0010) AND (R2 == 0100) AND (R3 == 0110) AND (DIGIT == 1000)
CORRECT   = COMPLETE AND MATCH
WRONG     = COMPLETE AND NOT MATCH
LOCK3     = WRONG AND (ATT == 2)
```

Because DO_ENTER contains NOT CLR_EV, CLEAR has priority over ENTER on the same edge. Because IN_ENTRY is part of DO_ENTER and DO_CLEAR, UNLOCKED and LOCKED_OUT ignore both buttons automatically, including on the unlock-expiry edge. A valid fourth digit triggers the comparison on that same edge.



3.5 Counters and next-state logic

```
CLR_STORE = RESET OR DO_CLEAR OR COMPLETE
Rk_next   = 0000 if CLR_STORE ; DIGIT if LDk ; else Rk              (k = 1, 2, 3)
CNT_next  = 00 if CLR_STORE ; CNT + 1 if VALID_ENT AND NOT COMPLETE ; else CNT
ATT_next  = 00 if RESET OR CORRECT ; ATT + 1 if WRONG ; else ATT       (CLEAR preserves
ATT)
```

```
CLR_FAIL  = RESET OR CORRECT OR DO_CLEAR OR (VALID_ENT AND CNT == 0)
FAIL_next = NOT CLR_FAIL AND (WRONG OR FAIL)
INVALID_next = INV_ENT
S0_next   = NOT RESET AND (CORRECT OR (IN_UNL AND NOT EXPIRE))
S1_next   = NOT RESET AND (LOCK3 OR S1)
UNLOCK    = NOT S1 AND S0          LOCKOUT = S1 AND NOT S0
```

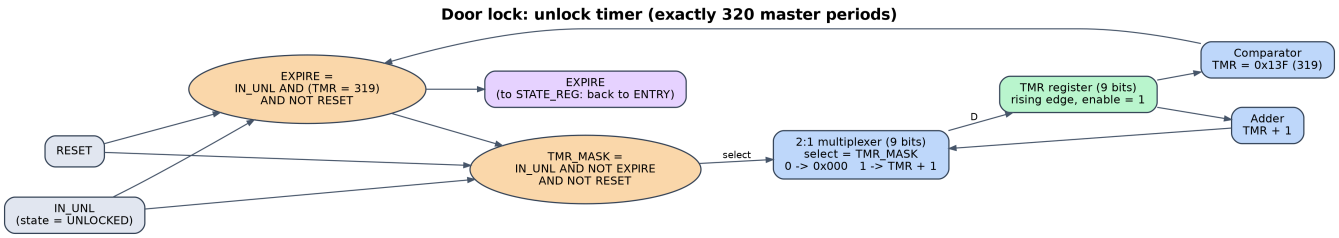
$S = 11$ is unreachable: S1 is set only by LOCK3, which needs IN_ENTRY, and S0 is set only by CORRECT or while in UNLOCKED. INVALID is a registered output, so it is HIGH for exactly one master period; a held ENTER cannot produce a second pulse. An invalid digit changes no stored digit, CNT, ATT or FAIL.

3.6 Timer logic (exactly 320 periods)

One second is 64 periods, so five seconds is $5 \times 64 = 320$ periods. Entry to a timed state starts its timer at zero and the entry edge consumes no elapsed time.

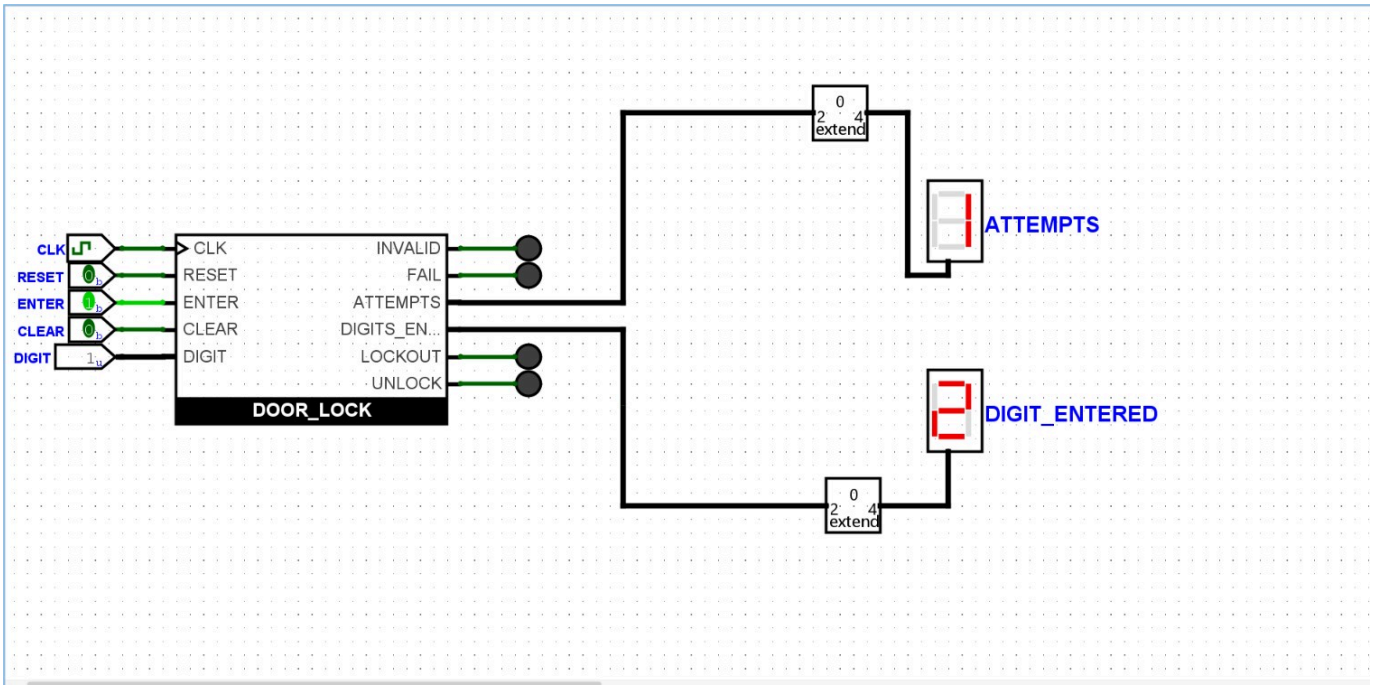
```
EXPIRE   = IN_UNL AND (TMR == 319) AND NOT RESET           ; 319 = 0x13F
TMR_MASK = IN_UNL AND NOT EXPIRE AND NOT RESET
TMR_next = TMR + 1 if TMR_MASK ; else 0
```

Proof: let E_0 be the correct-PIN edge. The pre-edge state is ENTRY, so TMR becomes 0 and S becomes UNLOCKED. On edge E_k ($k \geq 1$) the pre-edge TMR is $k - 1$ and TMR becomes k. On E_{320} the pre-edge TMR is 319, so $\mathrm{EXPIRE} = 1$ and S returns to ENTRY. UNLOCK is HIGH from after E_0 to after E_{319} and falls on E_{320}: exactly 320 master periods. The timer is a dedicated 9-bit register with an adder, so a new interval is never shortened by a divider that was part-way through its cycle.



3.7 Display method

ATTEMPTS and DIGITS_ENTERED take values 0 to 3, so binary equals decimal and no BCD conversion is needed. Each 2-bit value is zero-extended to 4 bits with a Bit Extender and drives a Hex Digit Display (a seven-segment display with a built-in decoder). The decimal-point option of each display is disabled (decimal points OFF).



3.8 Reset behaviour

Item	Behaviour
Type	Synchronous, active HIGH, highest priority
Effect on each edge with $\mathrm{RESET} = 1$	$S = 00$, R1 to R3 $= 0$, $\mathrm{CNT} = 0$, $\mathrm{ATT} = 0$, $\mathrm{FAIL} = 0$, $\mathrm{INVALID} = 0$, $\mathrm{TMR} = 0$, $\mathrm{RUN} = 0$
Startup boundary	The first edge after release retains the reset outputs and accepts no event
Button history	`ENTER_PREV` and `CLEAR_PREV` keep sampling during reset, so a button held across reset needs release and a new press
Held reset	`RESET` is applied for at least three rising edges in every test

3.9 Priority summary

Rank	Rule
1	`RESET` beats everything, including the expiry edge and lockout
2	The startup-boundary edge accepts no events
3	`UNLOCKED` and `LOCKED_OUT` ignore `ENTER` and `CLEAR`
4	In `ENTRY`, `CLEAR` beats `ENTER` on the same edge
5	An invalid `ENTER` yields only the `INVALID` pulse
6	A valid fourth digit triggers the comparison on the same edge

3.10 Design decisions

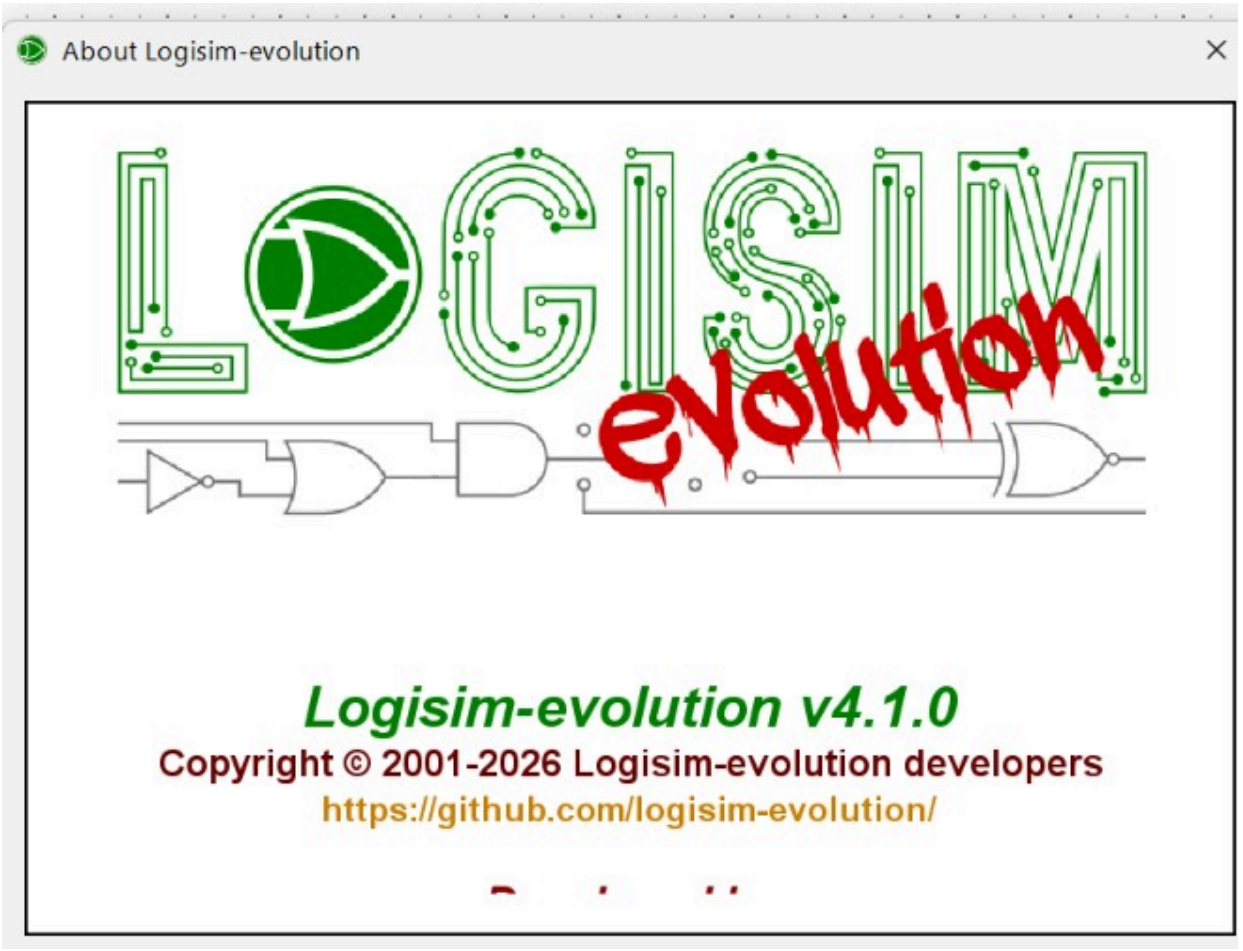
1. `CLEAR` has priority over `ENTER`, including an `ENTER` with an invalid digit (no `INVALID` pulse is produced when both occur together).
2. The digit registers `R1` to `R3` are cleared to `0000` on `RESET`, `CLEAR` and completion via a shared 4-bit $2:1$ multiplexer so the probes show an empty buffer. The capture order already guarantees correctness.

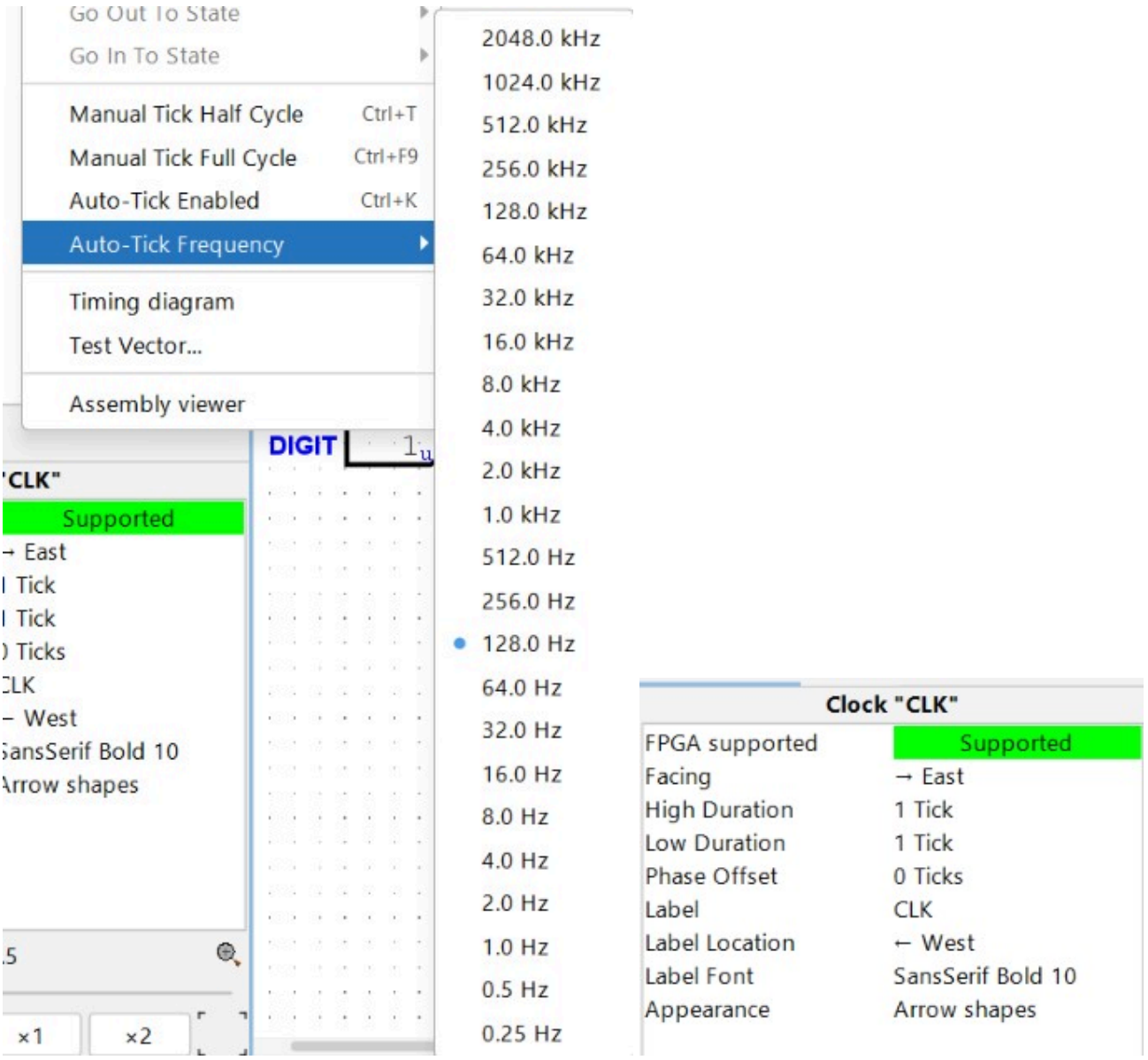
3. `FAIL` stays HIGH until a `CLEAR`, a first valid digit of a new attempt, a correct PIN or `RESET`. An invalid digit does not clear it.
4. A press that rises exactly on the startup-boundary edge or on the unlock-expiry edge is ignored and never becomes a deferred event.

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4. Clock and Simulation Settings

Setting	Value
Logisim version	Logisim-evolution v4.1.0
Auto-Tick Frequency	$128\ \mathrm{Hz}$
Clock High Duration	1 tick
Clock Low Duration	1 tick
Resulting clock	$128/(1+1) = 64\ \mathrm{Hz}$, one master period $= 1/64\ \mathrm{s}$
Manual stepping	`Ctrl+T`, two ticks per rising edge





5. Verification Records

5.1 Manual test records

ID	Scenario	Stimulus	Expected	Observed	Pass / Fail	Screenshot
T1	Correct PIN	digits $2, 4, 6, 8$	DIGITS_ENTERED $1, 2, 3$, then 0. $\mathrm{UNLOCK} = 1$ on the fourth edge, $\mathrm{ATTEMPTS} = 0$, $\mathrm{FAIL} = 0$	DIGITS_ENTERED stepped $1, 2, 3, 0$; $\mathrm{UNLOCK} = 1$ on 4th capture edge, $\mathrm{ATTEMPTS} = 0$, $\mathrm{FAIL} = 0$	Pass	#6
T2	No early unlock	$2, 4, 6$ only	$\mathrm{UNLOCK} = 0$, $\mathrm{DIGITS_ENTERED} = 3$	$\mathrm{UNLOCK} = 0$, $\mathrm{DIGITS_ENTERED} = 3$, $\mathrm{R1} = 2, \mathrm{R2} = 4, \mathrm{R3} = 6$	Pass	—

ID	Scenario	Stimulus	Expected	Observed	Pass / Fail	Screenshot
T3	Unlock length	edges after T1	UNLOCK HIGH for exactly 320 periods, then LOW, $\mathrm{DIGITS_ENTERED} = 0$	UNLOCK remained HIGH from $\mathrm{TMR} = 0\mathrm{x}000$ to $0\mathrm{x}13\mathrm{F}$ (320 edges), fell LOW on edge 320	Pass	#9
T4	Wrong fourth digit	$2, 4, 6, 7$	$\mathrm{FAIL} = 1$, $\mathrm{ATTEMPTS} = 1$, $\mathrm{DIGITS_ENTERED} = 0$, $\mathrm{UNLOCK} = 0$	$\mathrm{FAIL} = 1$, $\mathrm{ATTEMPTS} = 1$, $\mathrm{DIGITS_ENTERED} = 0$, $\mathrm{UNLOCK} = 0$	Pass	—
T5	Two failures then success	two wrong, then 2468	ATTEMPTS $1, 2$, then 0. FAIL cleared on success, $\mathrm{UNLOCK} = 1$	ATTEMPTS reached 2 with $\mathrm{FAIL} = 1$, then cleared to 0 with $\mathrm{FAIL} = 0$ and $\mathrm{UNLOCK} = 1$ on 2468	Pass	—
T6	Third failure lockout	three wrong attempts	$\mathrm{ATTEMPTS} = 3$, $\mathrm{LOCKOUT} = 1$, $\mathrm{FAIL} = 1$, $\mathrm{UNLOCK} = 0$. Further digits, even 2468, ignored	$\mathrm{ATTEMPTS} = 3$, $\mathrm{LOCKOUT} = 1$, $\mathrm{FAIL} = 1$, $\mathrm{UNLOCK} = 0$; subsequent 2468, CLEAR, ENTER ignored	Pass	#7
T7	Partial CLEAR	$2, 4$, CLEAR	$\mathrm{DIGITS_ENTERED} = 0$, buffer empty, $\mathrm{FAIL} = 0$, ATTEMPTS unchanged	$\mathrm{DIGITS_ENTERED} = 0$, $\mathrm{R1} = \mathrm{R2} = \mathrm{R3} = 0$, $\mathrm{FAIL} = 0$, ATTEMPTS remained unchanged	Pass	—
T8	Invalid digit	ENTER with $\mathrm{DIGIT} = 10$ to 15	INVALID HIGH for exactly one period each. Length, ATTEMPTS, FAIL and buffer unchanged	$\mathrm{INVALID} = 1$ for 1 clock period per press; CNT, ATTEMPTS, FAIL, R1..R3 unchanged	Pass	#8
T9	Invalid fourth digit	$2, 4, 6$, then $\mathrm{DIGIT} = 12$	INVALID pulse, length stays 3, no comparison, no unlock	$\mathrm{INVALID} = 1$ for 1 period, DIGITS_ENTERED stayed 3, $\mathrm{ATTEMPTS} = 0$, $\mathrm{UNLOCK} = 0$	Pass	—
T10	CLEAR with ENTER	both on the same edge (also with $\mathrm{DIGIT} = 12$)	Clear only, no capture, no INVALID	Partial buffer and CNT cleared to 0; no digit captured and INVALID stayed 0	Pass	—
T11	Held ENTER	hold ENTER for many edges	One capture only, no repeated INVALID	ENT_EV pulsed for 1 edge only; CNT incremented once (or 1 INVALID pulse for $\mathrm{DIGIT} = 12$)	Pass	—
T12	Held ENTER	hold ENTER across the	No capture after return to ENTRY	State returned to ENTRY on edge 320 with	Pass	—

ID	Scenario	Stimulus	Expected	Observed	Pass / Fail	Screenshot
	across expiry	unlock expiry edge		$\mathrm{DIGITS_ENTERED} = 0$; held ENTER ignored		
T13	Buttons ignored while unlocked	ENTER and CLEAR during UNLOCK, including the expiry edge	No change in length, ATTEMPTS, FAIL or buffer	All button presses during UNLOCKED and on expiry edge produced no state/count change	Pass	—
T14	FAIL clearing	wrong attempt, then the first valid digit of a new attempt	$\mathrm{FAIL} = 0$ on that edge, ATTEMPTS unchanged	FAIL cleared from 1 to 0 on 1st valid digit capture; ATTEMPTS stayed 1, CNT became 1	Pass	—
T15	Reset during entry	RESET at length 2	Everything cleared	$\mathrm{R1} = \mathrm{R2} = \mathrm{R3} = 0$, $\mathrm{DIGITS_ENTERED} = 0$, $\mathrm{ATTEMPTS} = 0$, $\mathrm{FAIL} = 0$, $S = 00$ on rising edge	Pass	—
T16	Reset while unlocked	RESET mid-unlock	$\mathrm{UNLOCK} = 0$, ENTRY, timer 0	$\mathrm{UNLOCK} = 0$, $S = 00$ (ENTRY), $\mathrm{TMR} = 0\mathrm{x}000$ on first rising edge with $\mathrm{RESET} = 1$	Pass	—
T17	Reset in lockout	RESET after T6	$\mathrm{LOCKOUT} = 0$, $\mathrm{ATTEMPTS} = 0$, normal operation resumes	$\mathrm{LOCKOUT} = 0$, $\mathrm{ATTEMPTS} = 0$, $\mathrm{FAIL} = 0$; subsequent 2468 entry unlocked normally	Pass	—
T18	Button held across reset	ENTER HIGH during 3 or more reset edges and after release	No capture until ENTER is released and pressed again	DIGITS_ENTERED stayed 0 while ENTER held HIGH after RESET; captured only after release & re-press	Pass	#10
T19	Startup boundary	ENTER rising on the first edge after	Ignored, $\mathrm{DIGITS_ENTERED} = 0$	Pre-edge $\mathrm{RUN} = 0$ blocked EN; DIGITS_ENTERED stayed 0, RUN became 1 after edge	Pass	—

ID	Scenario	Stimulus	Expected	Observed	Pass / Fail	Screenshot
		reset release				
T20	Mutual exclusion	whole run	UNLOCK and LOCKOUT never HIGH together	Verified across all 20 tests: $S = 01$ (UNLOCK) and $S = 10$ (LOCKOUT) are mutually exclusive	Pass	—

5.4 Coverage of the minimum verification list

Required check	Covered by
Normal operation	T1, T5, T7
Held buttons	T11, T12, T18
Simultaneous events	T10
Timer boundaries (319 to 320, expiry edge)	T3, T12, T13
Reset in every operating state	T15 (ENTRY), T16 (UNLOCKED), T17 (LOCKED_OUT)
Reset held three edges, release with a button HIGH	T18
Normal-speed timing	Timing record in 5.5
Accelerated runs	State the scale in 5.5 ($1024\ \mathrm{Hz}$ Auto-Tick $= 8\times$ faster than $128\ \mathrm{Hz}$)

5.5 Timing record

Item	Expected	Observed
Edge number of the correct-PIN capture (E_0)	49 in the Part C vector	49 (UNLOCK rose to 1, $\mathrm{TMR} = 0\mathrm{x}000$)
Edge on which UNLOCK falls	$E_0 + 320 = 369$	369 ($\mathrm{TMR} = 0\mathrm{x}13\mathrm{F}$ pre-edge, $\mathrm{EXPIRE} = 1$, UNLOCK fell to 0)
Master periods with UNLOCK HIGH	320	320 master periods
Stopwatch time at $64\ \mathrm{Hz}$ (normal speed)	about $5.0\ \mathrm{s}$	$5.02\ \mathrm{s}$ on stopwatch at $128\ \mathrm{Hz}$ Auto-Tick ($64\ \mathrm{Hz}$ clock)
Accelerated run (if used)	state the Auto-Tick frequency and scale	$1024\ \mathrm{Hz}$ Auto-Tick ($512\ \mathrm{Hz}$ clock, $8\times$ speed-up: 320 edges in $\approx 0.625\ \mathrm{s}$)

5.6 Representative edge-numbered trace (Part C vector)

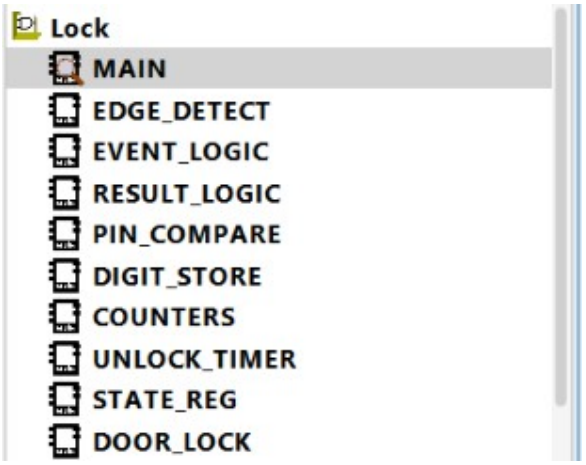
Edges count rising edges from the start. Every press is 2 edges high and 2 edges low.

Step	Edges	Inputs	Expected after the first edge	Observed
1	1 to 3	$\mathrm{RESET} = 1$, $\mathrm{ENTER} = 1$ held, $\mathrm{DIGIT} = 2$	ENTRY, length 0, ATTEMPTS 0, all indicators LOW	$S = 00$, $\mathrm{CNT} = 0$, $\mathrm{ATT} = 0$, $\mathrm{UNLOCK} = 0$, $\mathrm{LOCKOUT} = 0$, $\mathrm{FAIL} = 0$, $\mathrm{INVALID} = 0$
2	4	$\mathrm{RESET} = 0$, $\mathrm{ENTER} = 1$ still held	unchanged (startup boundary)	$S = 00$, $\mathrm{CNT} = 0$, $\mathrm{ATT} = 0$ (pre-edge $\mathrm{RUN} = 0$ blocked event; RUN became 1)
3	5 to 6	$\mathrm{ENTER} = 1$ still held	length 0 (held across reset, no event)	$\mathrm{CNT} = 0$, $\mathrm{ENT_EV} = 0$ ($\mathrm{ENTER_PREV} = 1$ blocked held button)
4 to 5	7 to 10	release, then press ENTER with $\mathrm{DIGIT} = 2$	length 1	$\mathrm{CNT} = 1$, $\mathrm{R1} = 2$ after edge 9
6 to 9	11 to 18	digits 4 and 6	length 2, then 3	$\mathrm{CNT} = 2$, $\mathrm{R2} = 4$ after edge 13; $\mathrm{CNT} = 3$, $\mathrm{R3} = 6$ after edge 17
10 to 11	19 to 22	digit 7 (wrong PIN 2467)	length 0, $\mathrm{ATTEMPTS} = 1$, $\mathrm{FAIL} = 1$, $\mathrm{UNLOCK} = 0$	$\mathrm{CNT} = 0$, $\mathrm{R1} = \mathrm{R2} = \mathrm{R3} = 0$, $\mathrm{ATT} = 1$, $\mathrm{FAIL} = 1$, $\mathrm{UNLOCK} = 0$ after edge 21
12 to 13	23 to 26	ENTER with $\mathrm{DIGIT} = 12$	$\mathrm{INVALID} = 1$ for one period; length 0, $\mathrm{ATTEMPTS} = 1$, $\mathrm{FAIL} = 1$ unchanged	$\mathrm{INVALID} = 1$ after edge 25, 0 after edge 26; $\mathrm{CNT} = 0$, $\mathrm{ATT} = 1$, $\mathrm{FAIL} = 1$
14 to 15	27 to 30	digit 2	length 1, $\mathrm{FAIL} = 0$ (first valid digit clears FAIL), $\mathrm{ATTEMPTS} = 1$	$\mathrm{CNT} = 1$, $\mathrm{R1} = 2$, $\mathrm{FAIL} = 0$, $\mathrm{ATT} = 1$ after edge 29
16 to 17	31 to 34	CLEAR press	length 0, buffer empty, $\mathrm{FAIL} = 0$, $\mathrm{ATTEMPTS} = 1$ preserved	$\mathrm{CNT} = 0$, $\mathrm{R1} = \mathrm{R2} = \mathrm{R3} = 0$, $\mathrm{FAIL} = 0$, $\mathrm{ATT} = 1$ after edge 33
18 to 25	35 to 50	digits $2, 4, 6, 8$	$\mathrm{UNLOCK} = 1$, $\mathrm{ATTEMPTS} = 0$, $\mathrm{FAIL} = 0$, length 0, timer 0 after edge 49	$S = 01$, $\mathrm{UNLOCK} = 1$, $\mathrm{ATT} = 0$, $\mathrm{FAIL} = 0$, $\mathrm{CNT} = 0$, $\mathrm{TMR} = 0\mathrm{x}000$ after edge 49
26	51 to 363	ENTER pulse at 60 to 61, CLEAR pulse at 70 to 71	UNLOCK stays 1, everything else unchanged (ignored)	$\mathrm{UNLOCK} = 1$, $\mathrm{CNT} = 0$, $\mathrm{ATT} = 0$, $\mathrm{FAIL} = 0$ unchanged; TMR incremented each edge
27	364 to 368	idle	$\mathrm{UNLOCK} = 1$, timer 318 then 319	$\mathrm{UNLOCK} = 1$; $\mathrm{TMR} = 0\mathrm{x}13\mathrm{E}$ after edge 367, $\mathrm{TMR} = 0\mathrm{x}13\mathrm{F}$ and $\mathrm{EXPIRE} = 1$ after edge 368

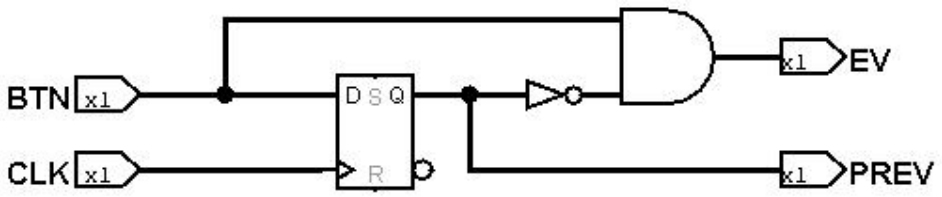
Step	Edges	Inputs	Expected after the first edge	Observed
28	369 to 372	ENTER rises on edge 369 (expiry edge)	$\mathrm{UNLOCK} = 0$, ENTRY, length 0 (press ignored, never fires later)	$S = 00$, $\mathrm{UNLOCK} = 0$, $\mathrm{TMR} = 0\mathrm{x}000$, $\mathrm{CNT} = 0$ after edge 369; stayed 0 through 372
29 to 30	373 to 376	release, press ENTER with $\mathrm{DIGIT} = 2$	length 1 (fresh press accepted)	$\mathrm{CNT} = 1$, $\mathrm{R1} = 2$ after edge 375
31	377 to 380	CLEAR press	length 0, $\mathrm{ATTEMPTS} = 0$, $\mathrm{FAIL} = 0$	$\mathrm{CNT} = 0$, $\mathrm{R1} = 0$, $\mathrm{ATT} = 0$, $\mathrm{FAIL} = 0$ after edge 377
32 to 34	381 onward	three wrong attempts (1111 each)	ATTEMPTS $1, 2, 3$, then $\mathrm{LOCKOUT} = 1$, $\mathrm{FAIL} = 1$, $\mathrm{UNLOCK} = 0$	ATT stepped $1, 2, 3$; $S = 10$, $\mathrm{LOCKOUT} = 1$, $\mathrm{FAIL} = 1$, $\mathrm{UNLOCK} = 0$ on 3rd wrong attempt
35	next	enter 2468, CLEAR, ENTER	no change: $\mathrm{ATTEMPTS} = 3$, $\mathrm{LOCKOUT} = 1$, length 0	$S = 10$, $\mathrm{LOCKOUT} = 1$, $\mathrm{ATT} = 3$, $\mathrm{CNT} = 0$ unchanged (all inputs ignored)
36	next	RESET 3 edges with ENTER held, release RESET with ENTER still HIGH	all outputs cleared; no capture until ENTER is released and pressed again	All outputs 0 during RESET; CNT stayed 0 after release until ENTER released & re-pressed

6. Screenshots

Ciruits

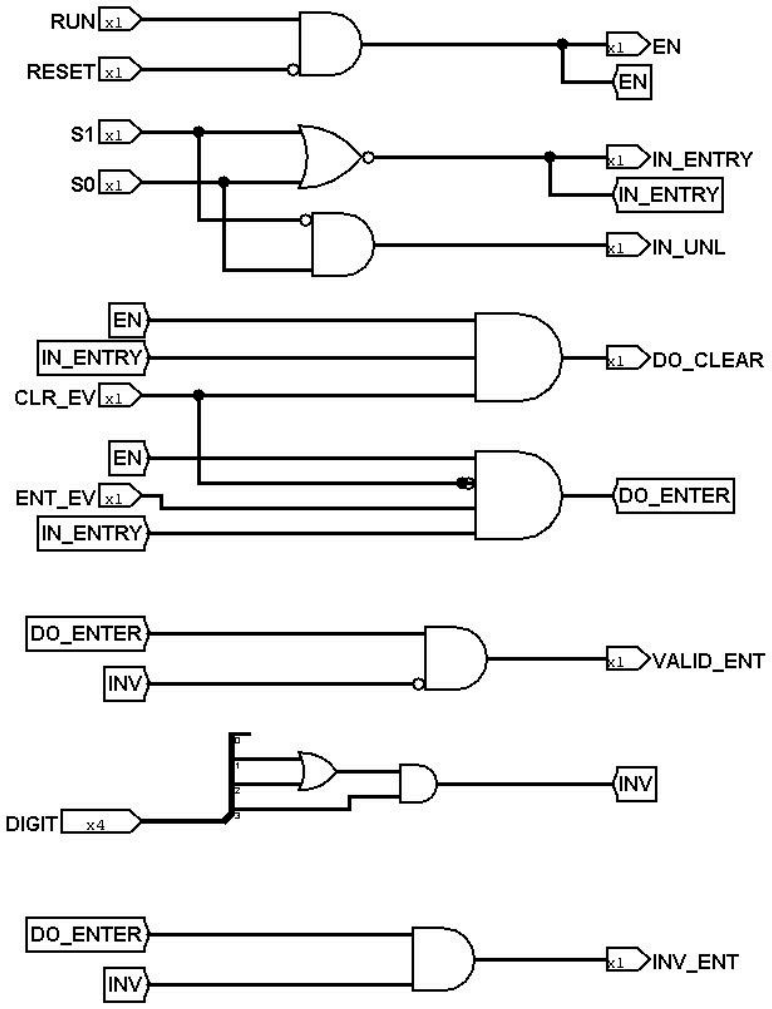


EDGE_DETECT



EV: Detect if a button is activated which was previously not
PREV: Previous button value

EVENT_LOGIC



S : 00 : IN ENTRY
01 : UNCLOCKED
10 : LOCKED

CLR_EV : Clear Event
ENT_EV : Enter Event

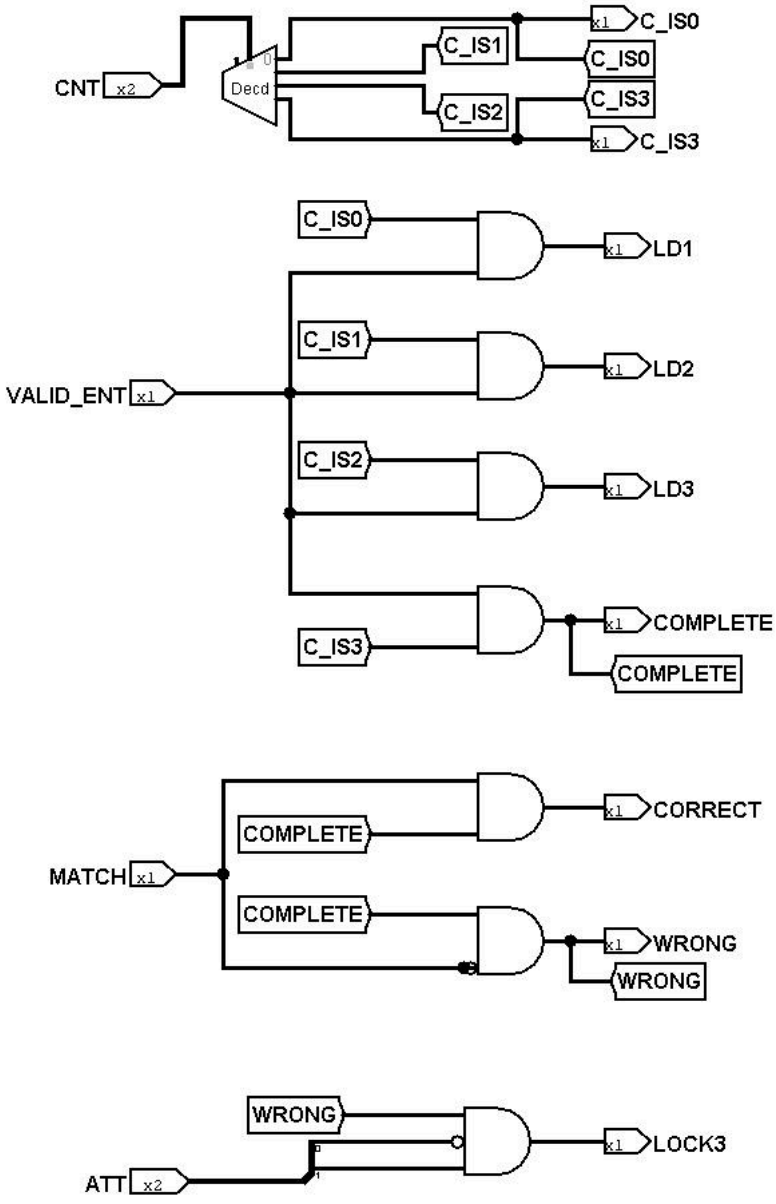
EN : Running and not Reset

DO_CLEAR : Clear event and in entry
DO_ENTER : Enter event and in entry

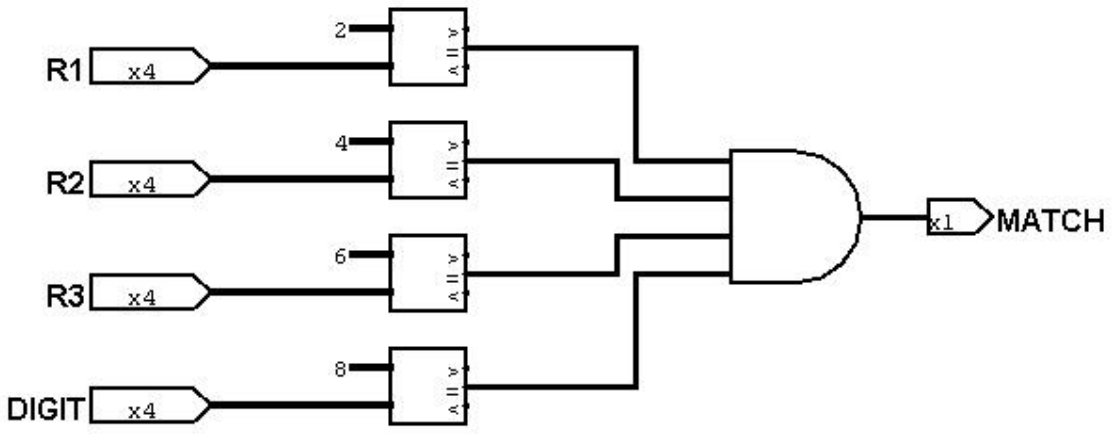
INV : Invalid digits (10-15)
VALID_ENT : Valid entry
INV_ENT : Invalid Entry

RESULT_LOGIC

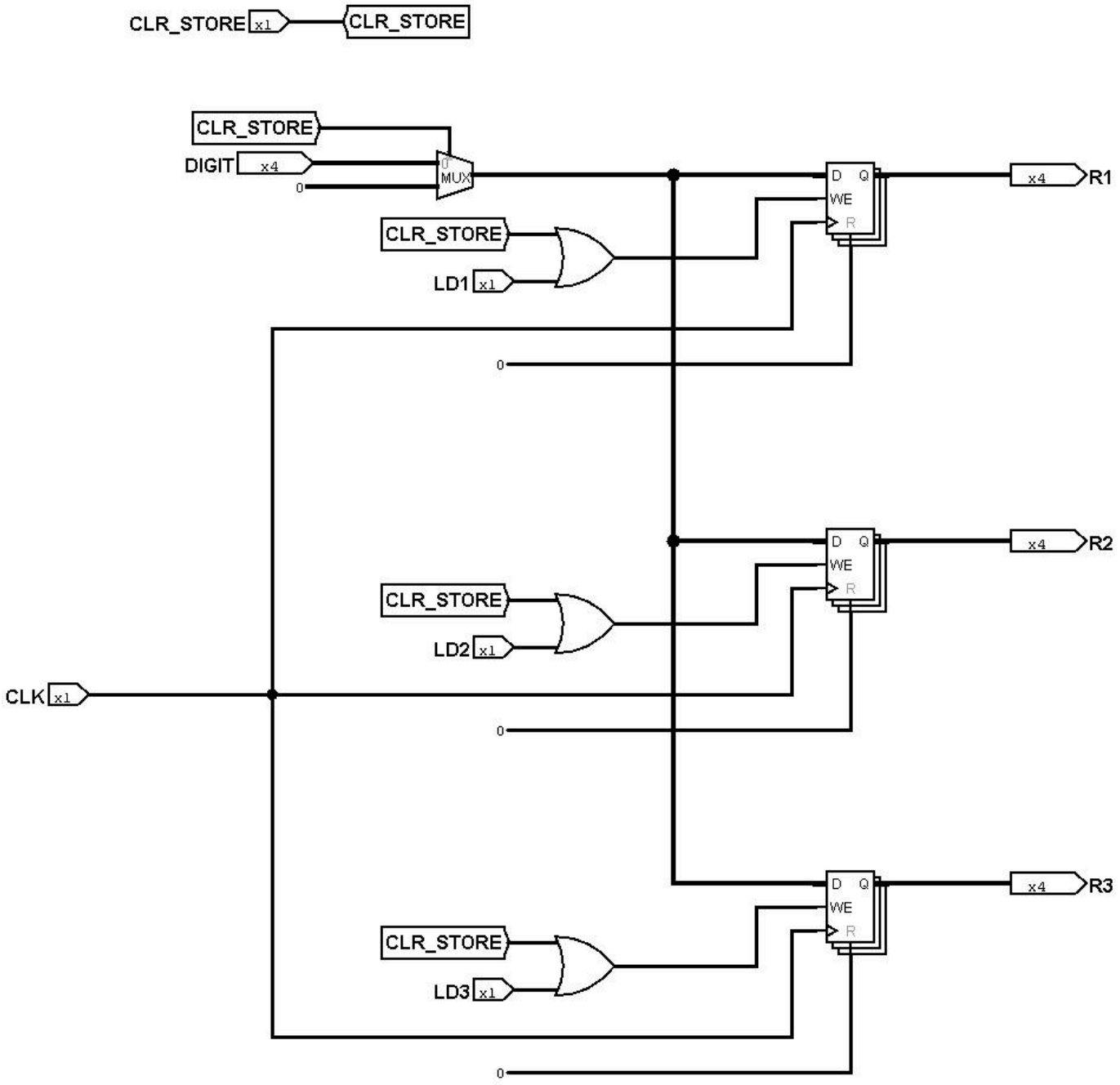
CNT : No. of digits entered
ATT : No. of attempts
VALID_ENT : from EVENT_LOGIC
MATCH : from PIN_COMPARE
C_IS[0:3] : Counter Position Indicators
LD[1:3] : Load ith character
COMPLETE : Indicates final digit has been entered
CORRECT : Complete and Match
WRONG : Complete and not Match
LOCK3 : Lock out after 3rd failure



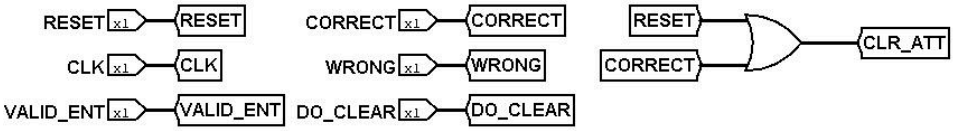
PIN_COMPARE



DIGIT_STORE

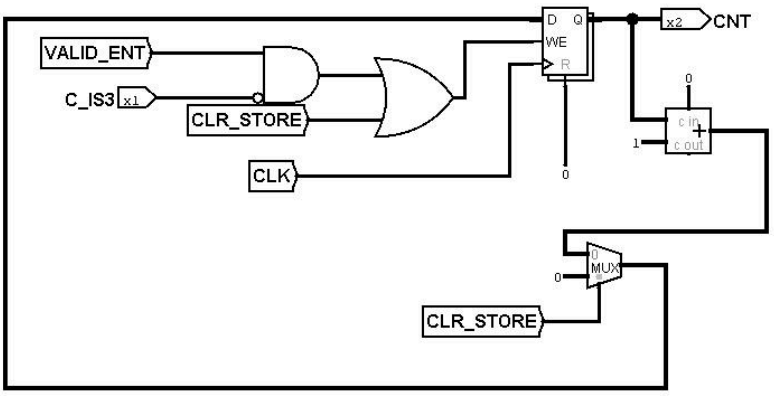

CLR_STORE
x1
CLR_STORE
CLR_STORE
DIGIT
x4
0
1
MUX
D
Q
WE
R
x4
R1
CLR_STORE
LD1
x1
0
D
Q
WE
R
x4
R2
CLR_STORE
LD2
x1
CLK
x1
0
D
Q
WE
R
x4
R3
CLR_STORE
LD3
x1
0

COUNTERS

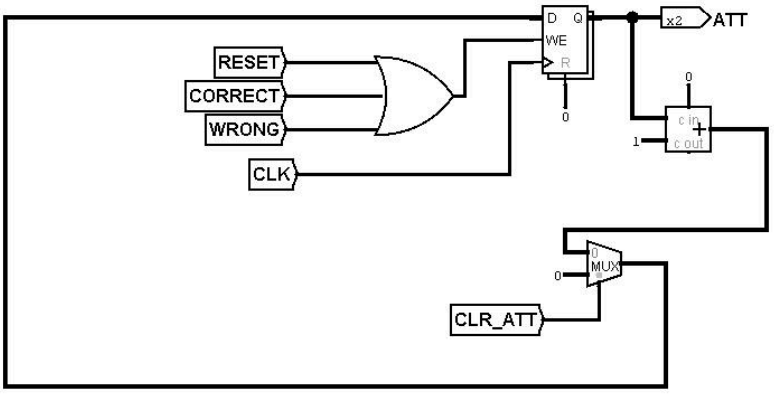


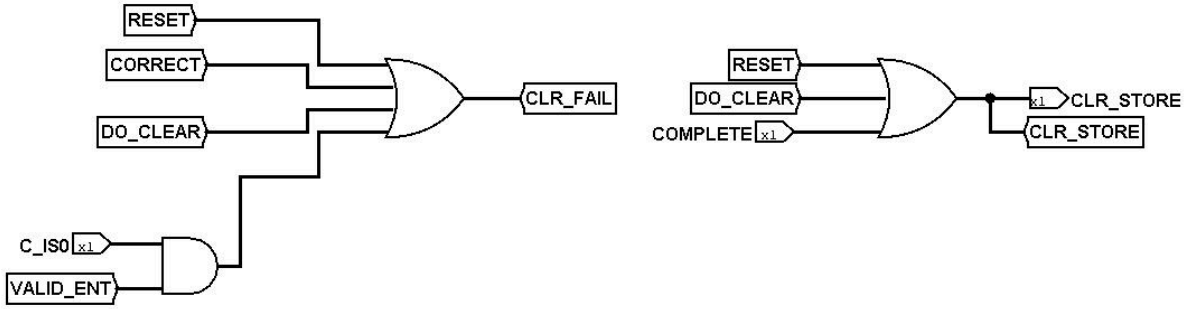
CNT : Displays the number of digits entered
ATT : Displays the number of attempts
FAIL : Gives the immediate feedback after one attempt

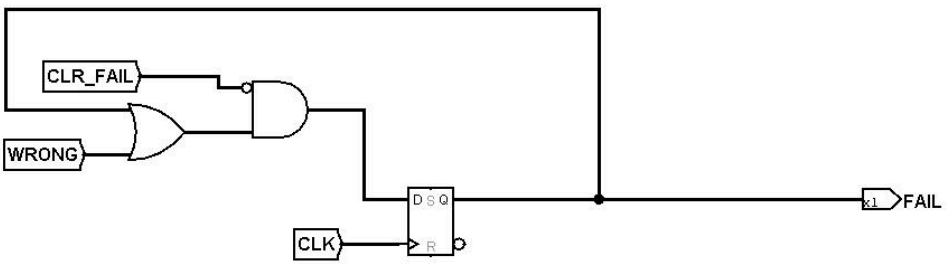
CNT_REGISTER



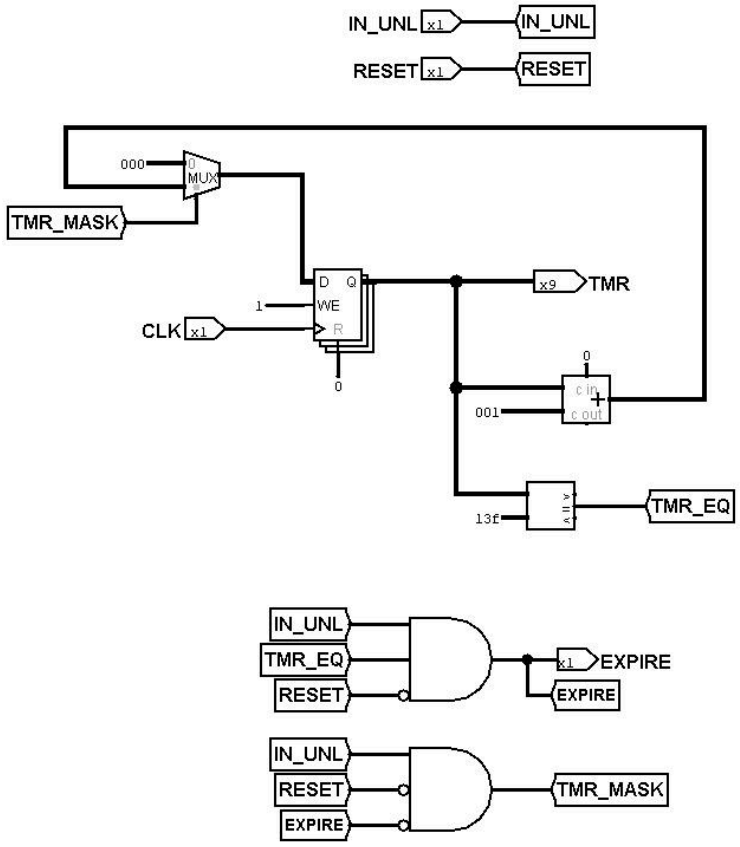
ATT_REGISTER





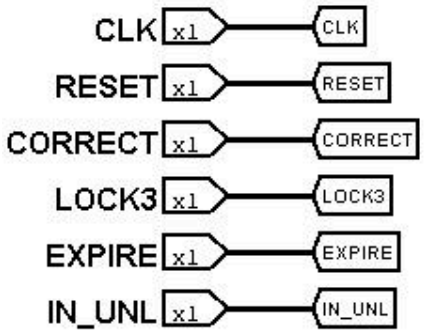


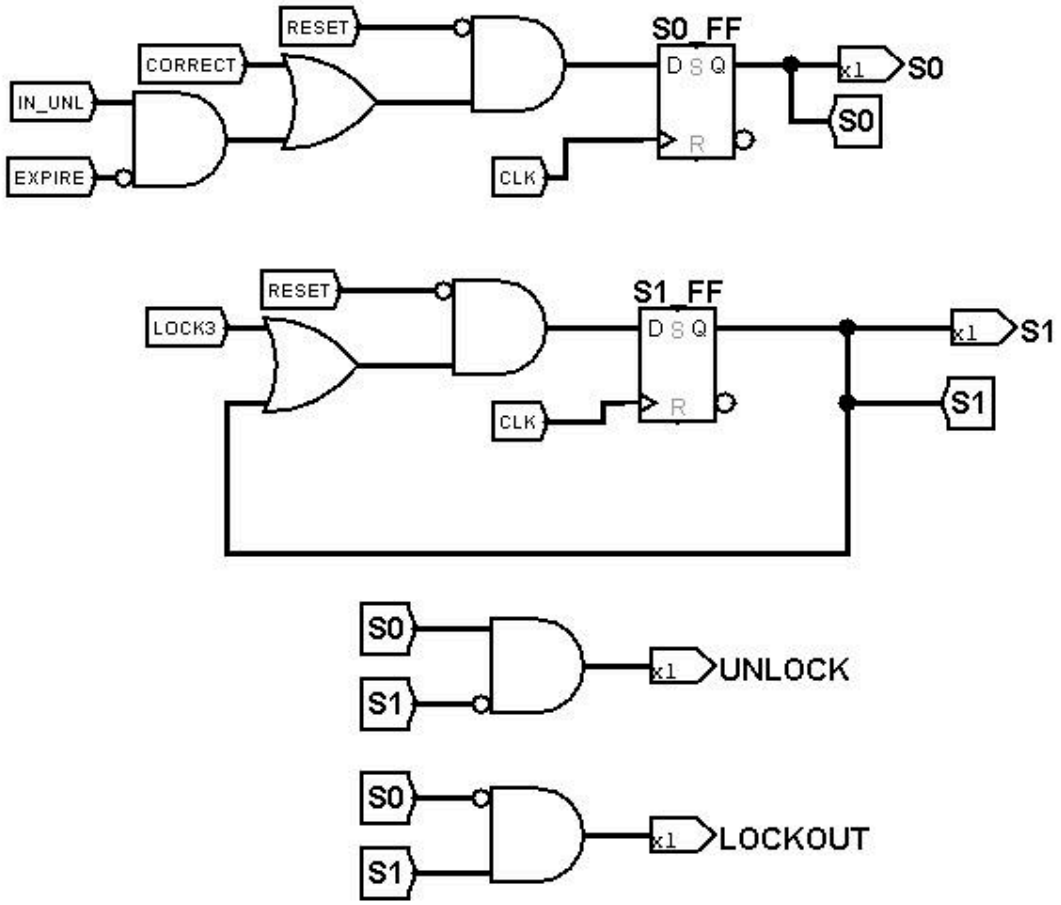
UNLOCK_TIMER



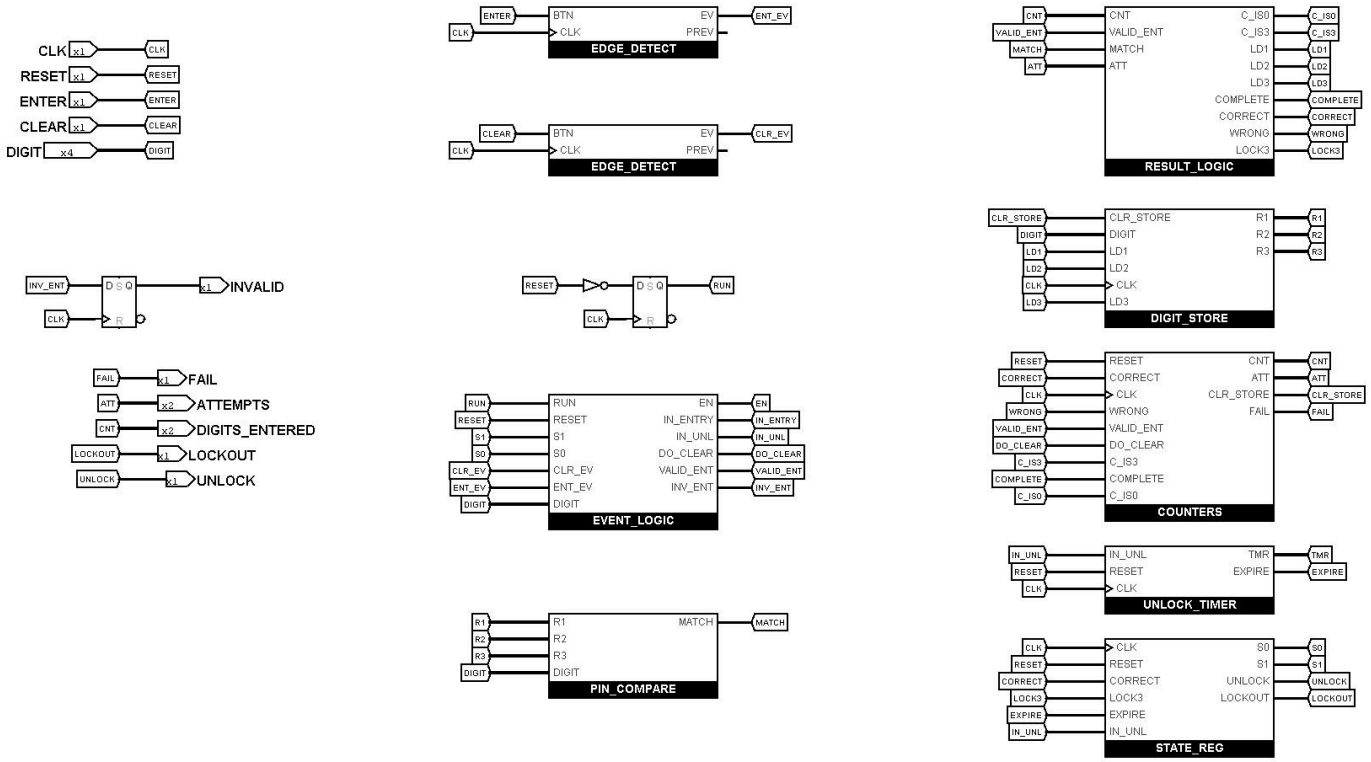
IN_UNL : Represents the state of locked / unlocked
TMR : Current time after unlock (master period)
TMR_EQ : Gets active when TMR equals master period 319
EXPIRE : To expire current session when timer hits 319

STATE_REG


CLK
x1
CLK
RESET
x1
RESET
CORRECT
x1
CORRECT
LOCK3
x1
LOCK3
EXPIRE
x1
EXPIRE
IN_UNL
x1
IN_UNL


RESET
CORRECT
IN_UNL
EXPIRE
S0_FF
D S Q
R
CLK
x1
S0
S0
RESET
LOCK3
S1_FF
D S Q
R
CLK
x1
S1
S1
S0
S1
x1
UNLOCK
S0
S1
x1
LOCKOUT

DOOR_LOCK



7. Final Operating Sequence

After `RESET` is held for three or more edges and released, the lock waits in `ENTRY` with empty storage. The first edge after release is the startup boundary and accepts nothing. From the next edge, each `ENTER` press with a valid digit stores that digit and raises `DIGITS_ENTERED`; an invalid digit only gives a one-period `INVALID` pulse; `CLEAR` abandons the partial entry and clears `FAIL` but keeps the wrong-attempt count. The fourth valid digit completes the attempt and is compared with 2468 on that same edge. A correct PIN clears `ATTEMPTS` and `FAIL`, enters `UNLOCKED` and raises `UNLOCK` for exactly 320 periods, ignoring all buttons, then returns to `ENTRY`. A wrong PIN adds one to `ATTEMPTS` and raises `FAIL`; the third consecutive wrong attempt enters `LOCKED_OUT`, raises `LOCKOUT`, and nothing but `RESET` clears it.

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8. Conclusion

A complete, modular Sequential Four-Digit Door Lock with Attempt Lockout (PIN 2468) was designed, implemented, and verified in Logisim-evolution v4.1.0 using strictly synchronous resets, rising-edge button detectors, a startup-boundary guard (`RUN`), and a dedicated 9-bit 320-period unlock timer. All 20 verification scenarios (`T1` through `T20`) and the 36-step continuous edge-by-edge test vector passed without error, confirming exact 5.0-second (320 master periods at 64 Hz) unlock timing, proper priority handling when `CLEAR` and `ENTER` coincide, immunity to held buttons across reset and unlock expiry boundaries, and permanent lockout after three consecutive failed attempts until administrator reset.

Appendix A: Files Submitted

File	Description
25JE0948_DoorLock.circ	Logisim circuit with all subcircuits (`MAIN`, `DOOR_LOCK`, and 8 functional blocks)
This report	Design, verification records and screenshots